

SEMESTER 5

Object Oriented Design Using UML

Course Code	HN C ST509	CIE Marks	40
Teaching Hours/Week (L:T:P)	3:1:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. To master the fundamental principles of object-oriented methodology—including abstraction, encapsulation, and inheritance—and apply analysis techniques to model the static and dynamic behavior of software systems.
2. To implement rigorous system and object design strategies and utilize the Unified Modeling Language (UML) along with standard design patterns to construct robust, scalable software architectures.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Object Oriented Development - Modeling Concepts–Object Oriented Methodology. Object Oriented Themes - Abstraction - Encapsulation - Combining Data and Behavior - Sharing - Emphasis on Object Structure. Object Oriented Models. Object modeling: Objects and Classes, Links and Associations, Advanced links and Association Concepts, Generalization and Inheritance, Grouping Constructs, A Sample Object Model.	11
2	Dynamic modeling: Events and States, Operations, Nested state diagrams, Concurrency, Advanced Dynamic Modeling Concepts, A sample Dynamic Model, Relationship of Object and Dynamic models. Functional modeling: Functional models, Data Flow Diagrams, Specifying Operations, Constraints, A sample Functional Model. Analysis: Object Modeling - Identifying Object Classes - Preparing a Data Dictionary - Identifying Associations. Dynamic Modeling - Preparing a Scenario - Interface Format - Identifying Event - Building a State Diagram. Functional Modeling - Identifying input and Output Values - Building Data Flow Diagram-Describing Functions.	11

3	System Design: Breaking System into Subsystems, Identifying Concurrency, Allocating Subsystems to Processors and Tasks, Managing Data Stores, Handling of Global Resources, Common Architectural Framework. Object Design: Overview of Object design, Combining the three models, Designing algorithms, Design optimization, Implementation of control, Adjustment of inheritance, Design of association, Object representation, Physical packaging. Documenting design decisions - Comparison of methodologies.	11
4	Unified Modeling Language (UML)-UML introduction & benefits-Different types of UML diagrams- Behavioral. Diagrams-Activity Diagram - Use Case Diagram - State Machine Diagram - Sequence Diagram , structural diagrams.- Class Diagram-Object Diagram-Component Diagram-Composite Structure Diagram-Deployment Diagram-Package Diagram. UML tools and their needs. Design Patterns-Creational - Structural – Behavioral.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop object models that represent the static structure of a real-world system by utilizing concepts such as classes, associations, generalization, and inheritance.	Apply
CO2	Develop dynamic and functional models using state transition diagrams and data flow diagrams to capture system events, concurrency, and operational requirements.	Apply
CO3	Translate analysis models into detailed system and object designs by defining subsystems, optimizing algorithms, and managing data storage strategies.	Apply
CO4	Solve real time problems using various Modeling concepts for managing projects in multidisciplinary environments	Apply

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3								3
CO2	3	3	3	3							3
CO3	3	3	3	3							3
CO4	3	3	3	3							3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Object Oriented Modeling and Design	James Rumbaugh	Prentice Hall India	1/e
2	Object Oriented Analysis and Design with Applications	Grady Booch	Pearson Education Asia References	3/e
3	UML Distilled: A Brief Guide to the Standard Object Modeling Language	Martin Fowler	Addison-Wesley Professional	3/e

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Object Oriented Software Engineering	Ivan Jacobson	Pearson Education Asia.	3/e
2	Object Oriented Software	Berno Bruegge, Allen H.	Pearson Education	3/e

	Engineering	Dutoit	Asia.	
3	Object Oriented Analysis and Design using UML	H. Srimathi, H. Sriram, A. Krishnamoorthy	Scitech Publications.	1/e
4	UML and C++ practical guide to Object Oriented development	Richard C.Lee& William	Prentice Hall India	2/e.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/106105224

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
FIFTH SEMESTER B. TECH HONOURS DEGREE EXAMINATION, MONTH AND YEAR				
Course Code:				
Course Name:				
Max. Marks: 60			Duration: 2 hours 30 minutes	
	PART A			
		Answer all questions. Each question carries 3 marks	CO	Marks
1		List and define the three models in object oriented methodology.	1	(3)
2		Differentiate aggregation and composition in object oriented methodology with example.	1	(3)
3		Define the terms event and state in the context of dynamic modelling.	2	(3)
4		Describe the concept of interface format in dynamic modeling.	2	(3)
5		What are the adjustments that can be done to increase the chance of inheritance?	3	(3)
6		What are the factors to be considered for the design of the algorithm by a designer?	3	(3)
7		Explain Unified Modeling Language (UML).	4	(3)
8		Mention the purpose of use case diagram.	4	(3)
PART B				
Answer any one full question from each module. Each question carries 9 marks				
Module 1				
9	a)	What is object oriented methodology? Explain the stages involved in the object oriented methodology.	1	14
10	a)	Explain the different Object-Oriented Themes.	1	14
Module 2				
11	a)	Describe process, data flow and datastore used in data flow diagrams with example.	2	14

12	a)	Explain the concept of dynamic modeling and the steps performed for constructing a dynamic model.	2	14
Module 3				
13	a)	How the design optimization is performed?	3	7
	b)	Explain the concept of documenting design decisions in object design and its importance for maintaining consistency, understanding, and future modifications.	3	7
14	a)	Explain physical packaging in object design.	3	7
	b)	What are the three basic approaches to implementing the dynamic model?	3	7
Module 4				
15	a)	Explain the Unified Modeling Language (UML) and its role in object-oriented analysis and design.	4	14
16	a)	Explain the different types of design patterns.	4	14
